INCH-POUND
A-A-51187C
April 15, 2020
SUPERSEDING
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July 27, 2001

COMMERCIAL ITEM DESCRIPTION

WELDING TORCHES AND COMPONENT PARTS, GAS TUNGSTEN ARC WELDING (GTAW)

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers gas tungsten arc welding (GTAW) torches, also referred to as tungsten inert gas (TIG) welding torches.

2. CLASSIFICATION. The gas tungsten arc welding torches shall be of the following types, classes, styles, and sizes. The type, class, style, and size to be furnished shall be as specified (see 7.3(b)).

Type I - Air cooled
Type II - Water cooled

Class 1 – Rigid neck
Class 2 – Flexible neck

Style A – 60 to 75 degrees
Style B – 90 degrees
Style C – 180 degrees (pencil style)

Size - Sizes are shown in tables I and II

TABLE I. Sizes of type I air-cooled GTAW torches.

Size	Current Rating (A)
080	80
100	100
125	125
150	150
200	200

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AMSC N/A FSC 3431

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TABLE II. Sizes of type II water-cooled GTAW torches.

Size	Current Rating (A)
180	180
200	200
250	250
300	300
350	350
500	500

3. SALIENT CHARACTERISTICS.

3.1 General Requirements. The torch shall be provided complete with fittings and ready for use when connected to a gas source hose, power source cable, and, if a type II torch, water source hoses. Unless otherwise specified (see 7.3(c)), torches manufactured to metric dimensions shall be considered if they meet dimensional specifications, cable and hose connector standards, and other requirements stated herein. Electrodes, back caps, collets, collet bodies, gas cups (also called nozzles), and gas lenses shall be provided either as individual items or prepackaged in multiple-size parts kits as specified herein. In either case, the collets, collet bodies, gas cups, and gas lenses shall be compatible with electrode diameters, and the back caps compatible with electrode lengths specified herein.

3.2 Components. The torch shall include, as a minimum, the following components:

3.2.1 Torch body and handle. The current carrying components of the torch shall be made of high-conductive copper. The torch body and handle shall be made of lightweight, high-impact resistant, nonconductive materials.

3.2.2 Gas valve. When specified (see 7.3(d)), the torch body shall have an integral gas valve to provide on-off control of the inert gas flow.

3.2.3 Electrode(s) and back cap(s). If required, the electrode(s) shall have a diameter(s) of 0.020, 0.040, 1/16, 3/32, 1/8, 5/32, 3/16, or 1/4 inch and a length(s) of 2, 3, or 7 inches as specified (see 7.3(e)). If required, the back cap length(s) shall be short, medium, or long for compatibility with an electrode(s) of 2, 3, or 7 inches, respectively, as specified (see 7.3(f)).

3.2.4 Collet(s). If required, a collet(s) shall be provided to fit the electrode(s) selected in 3.2.3 as specified (see 7.3(g)).

3.2.5 Collet body. If required, a collet body shall be provided for each collet size or a range of collet sizes as specified (see 7.3(h)). The discharge pressure of the regulator shall be preset to a level that is compatible with the pressure rating of the flowmeter.

3.2.6 Gas cup(s). If required, a gas cup(s) (also known as a nozzle) with a diameter(s) of 3/16, 1/4, 5/16, 3/8, 7/16, 1/2, 5/8, 11/16, or 3/4 inch shall be provided as specified (see 7.3(i)). Unless

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otherwise specified (see 7.3(j)), the length of the gas cup(s) shall be in the range of 1 to 2 inches. Unless otherwise specified (see 7.3(k)), the gas cup(s) shall be made of alumina.

3.2.7 Gas lens(es). When specified (see 7.3(l)), the gas lens(es) shall be provided for the collet size(s) selected in 3.2.4. Unless otherwise specified (see 7.3(m)), the gas lens(es) provided shall be that considered by the manufacturer as standard.

3.2.8 Parts kit. When specified (see 7.3(n)), the back cap, electrodes, collets, collet bodies, and gas cups shall be provided as a parts kit.

3.2.9 Power cable, gas hose, and water hoses. Unless otherwise specified (see 7.3(o)), the type I torch shall be assembled with a 12.5-foot length of power cable and gas hose. Unless otherwise specified (see 7.3(p)), the type II torch shall be assembled with a 12.5-foot length of power cable, gas hose, and water hoses. The gas and water hoses shall have a minimum pressure rating of 150 pounds per square inch gage. The gas hose shall conform to the requirements of the Association for Rubber Products Manufacturers (ARPM) IP-7, "Specification for Rubber Welding Hose", for welding hoses and have fittings which conform to the requirements of Compressed Gas Association (CGA) E-1, "Standard for Rubber Welding Hose and Hose Connections for Gas Welding, Cutting, and Allied Processes", connection number 032.

3.2.10 Cable and hose cover. When specified (see 7.3(q)), a zippered cable and hose cover shall be provided to protect the cable and hose assembly.

3.2.11 Tools. Special tools for replacement or exchange of torch components shall be provided.

3.2.12 Optional accessories. Optional accessories shall be furnished as specified (see 7.3(r)).

3.3 Performance. The type I torch shall be rated at not less than a 60-percent duty cycle, and the type II torch shall be rated at a 100-percent duty cycle. A duty cycle is defined as the ratio of arc time to total time. The total time of one complete test cycle shall be 10 minutes.

3.4 Torch body marking. Unless otherwise specified (see 7.3(s)), the torch body shall be permanently identified with the manufacturer's name or trademark and the manufacturer's model or part number.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

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5.2 Market Acceptability. The product offered must have been previously sold either to the government or on the commercial market.

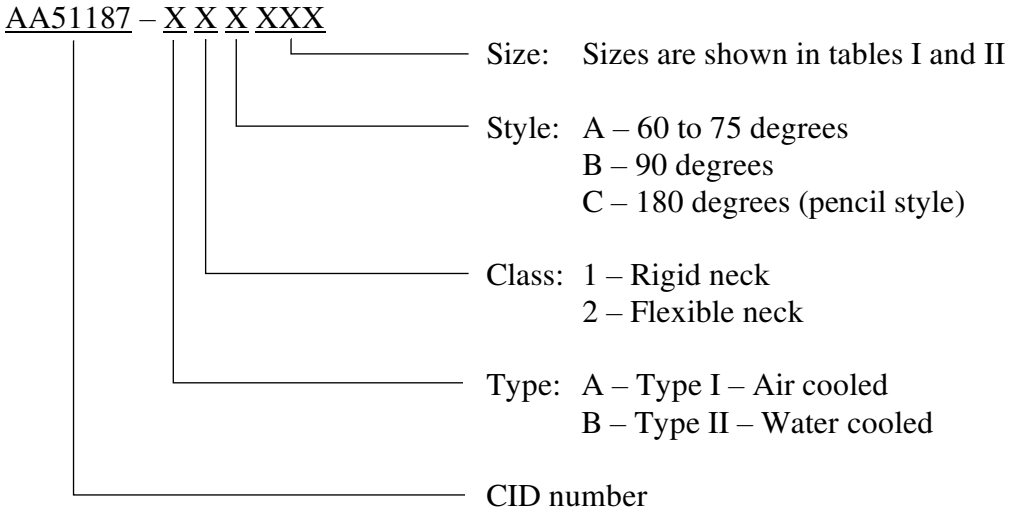
5.3 Inspection requirements. The torch shall be inspected to determine compliance with all requirements specified in this CID.

5.4 Warranty. Unless otherwise specified (see 7.3(t)), the manufacturer's standard commercial warranty, terms, and conditions shall apply.

6. PACKAGING. Unless otherwise specified (see 7.3(u)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. Copies of the FAR may be obtained from https://www.acquisition.gov/far/.

7.2.2 CGA standards. Copies of CGA standards may be obtained from the Compressed Gas Association, https://www.cganet.com/.

7.2.3 ARPM standards. Copies of ARPM standards may be obtained from the Association for Rubber Products Manufacturers, https://arpminc.com/.

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7.3 Ordering data. Purchaser should specify the following:

(a) CID title, number, date, and CID PIN.
(b) Type, class, style, and size as required (see 2).
(c) Torch dimensions, if different (see 3.1).
(d) Gas valve, if required (see 3.2.2).
(e) Electrode diameter(s) and length(s), as specified (see 3.2.3).
(f) Back cap length(s), as specified (see 3.2.3).
(g) Collet(s), as specified (see 3.2.4).
(h) Collet body, as specified (see 3.2.5).
(i) Gas cup diameter(s), as specified (see 3.2.6).
(j) Gas cup length, if different (see 3.2.6).
(k) Gas cup material, if different (see 3.2.6).
(l) Gas lens size(s), if required (see 3.2.7).
(m) Gas lens(es), if different (see 3.2.7).
(n) Parts kit, if required (see 3.2.8).
(o) Type I torch power cable and gas hose assembly, if different (see 3.2.9).
(p) Type II torch power cable, gas hose, and water hose assembly, if different (see 3.2.9).
(q) Cable and hose cover, if required (see 3.2.10).
(r) Optional accessories, as specified (see 3.2.12).
(s) Torch body marking, if different (see 3.4).
(t) Warranty, if different (see 5.4).
(u) Packaging requirements, if different (see 6).

MILITARY INTERESTS:

Custodians:
Army-AR
Navy-SH
DLA-GS

Reviewer:
Army-CE, CR4
Navy-MC, YD

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
DLA-GS6

(Project 3431-2020-003)

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